

Baber Rehman

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SUMMARY

I wish to play the leading role in the well-being of the humanity. Love to explore what's happening in the box instead of seeing things as a black box. Have a quality to work and adjust in dynamic environments.

ACADEMICS

2018 - 2023 PhD in Computer Science at **University of Hong Kong**, Hong Kong

Research Area: Programming Languages, Type Systems
Functional Programming, Object-Oriented Programming

Supervisor: [Dr. Bruno C. d. S. Oliveira](#)

2011 - 2015 Bachelor's Degree at **University of Punjab**, Lahore, Pakistan (CGPA: 3.81/4.0)

Major: Computer Science

Result: Distinction (Class: 1/54, Batch: 2/191)

RESEARCH SUMMARY

Programming languages define the constructs to instruct the machines. Various programming languages paradigms provide disparate styles of talking to the machines, such as imperative programming and functional programming. It has become essential with the growing need of technological transformation and pressing automation to design user-oriented, accessible, and robust programming languages. The more errors a programming language captures before compilation, the robust the program is. Such a language provides some static guarantees that the programs may not terminate abnormally at runtime. The study of type systems is an area of practical research interest to build robust programs. This is reflected by the fact that *a sound type system guarantees the absence of certain (abnormal) program behaviors*.

My current research revolves around type systems. Specifically, I am working on the integration of intersection types and union types. Intersection and union types are powerful features available in many modern programming languages such as Scala, Typescript, and Ceylon. The adequacy of intersection and union types lies in the fact that they naturally are able to encode various advance programming constructs, for example, function overloading, multiple interface inheritance, and nested composition.

TECHNICAL SKILLS

Golang

Scala

Haskell

Chef / Ansible

Docker / Kubernetes

Micro-services / API's

Bash / Python / Ruby

Coq Theorem Prover

Amazon Web Service (EC2, VPC, S3, RDS)

Microsoft Azure (VNet, Virtual Machines)

Computer Networking / VPC / VPN / PfSense

Jenkins / Git

Mongo DB

PROFESSIONAL EXPERIENCE

Compiler Engineer

Nov 2023 - Present

Huawei, Hong Kong Research Center

- At Huawei's Hong Kong Research Center, I work with a team where we design languages and compilers that let developers talk to hardware more effectively. I started by creating a P4-inspired DSL for programmable networking—giving engineers a high-level way to describe packet-processing logic that compiles to efficient data-plane code.
- Now I'm applying similar principles to AI acceleration, working across TileLang, MLIR (AscendNPU-IR), and LLVM-IR to build compiler stacks that map deep-learning workloads optimally onto Huawei's Ascend NPUs. My goal is to bridge the abstraction gap: turning domain-specific intent into fast, portable machine code.

Resident Tutor

Sep 2019 - Dec 2023

New College, The University of Hong Kong

- Main ethos of New College are Sustainability and Responsibility
- Maintain a lively environment for the students
- Assist students both academically and socially
- Mental Health First Aid Certification (Hong Kong)
- Suicide Prevention Service (Hong Kong)
- Responsible for the Photography Team, Web Team, and News and Media Team

Teacher Assistant

Sep 2018 - Jun 2019

The University of Hong Kong

- Functional Programming
- Principal of Programming Languages

Software Engineer (DevOps) – Team Lead

June 2015 - Aug 2018

CloudPlex PVT LTD, Lahore, Pakistan

- Build a cloud platform to automate cloud orchestration and deployment
- Worked with AWS and Azure cloud services including VPC, S3, EC2, VNet
- Worked on automating the configuration of firewalls (PfSense and Fortinet)
- Lead one local team in Lahore and one remote team located in Argentina
- Conducted technical interviews of candidates for the backend team
- Held meetings with clients for requirement gathering, demos, and delivery
- Tools and technologies: Golang, Chef, Ansible, Networking, Docker, Shell scripting, Cloud Computing

Visiting lecturer

Oct 2015 - Jun 2016

University of the Punjab, Lahore, Pakistan

- Fall 2015: Computer organization and Assembly Language course
- Spring 2016: Enterprise Application Development course (C#/MVC4)

Intern

Jul 2014 - Sep 2014

Mentor Graphics, Lahore, Pakistan

- Worked in Java programming language and used eclipse as a tool
- Explored Java GUI library i.e. SWT
- Did debugging of source code of Sourcery Codebench (an eclipse based IDE used in Mentor Graphics) and removed few of its GUI bugs

Teacher Assistant

Oct 2013 - Jul 2015

University of the Punjab, Lahore, Pakistan

- Discrete Mathematics
- Data Structures and Algorithms
- Database Systems
- System Programming
- Enterprise Application Development

Research Assistant

Jun 2013 - Oct 2013

AIM Lab, University of the Punjab, Lahore, Pakistan

- Worked with business process management research repository

CONFERENCE PROGRAM COMMITTEES

- ❖ Program Committee ECOOP 2026
- ❖ Artifact Evaluation Committee OOPSLA 2026
- ❖ Artifact Evaluation Committee POPL 2026
- ❖ Program Committee OOPSLA 2025
- ❖ Artifact Evaluation Committee ECOOP 2025
- ❖ Artifact Evaluation Committee PLDI 2025
- ❖ Artifact Evaluation Committee ISSTA 2024
- ❖ Artifact Evaluation Committee ECOOP 2024
- ❖ Extended Review Committee ECOOP 2024
- ❖ Artifact Evaluation Committee PLDI 2024
- ❖ Artifact Evaluation Committee POPL 2024
- ❖ Artifact Evaluation Committee ATC 2023
- ❖ Artifact Evaluation Committee OSDI 2023
- ❖ Artifact Evaluation Committee ECOOP 2023
- ❖ Extended Review Committee ECOOP 2023
- ❖ Artifact Evaluation Committee POPL 2023
- ❖ Artifact Evaluation Committee APLAS 2022
- ❖ Artifact Evaluation Committee ECOOP 2022
- ❖ Extended Review Committee ECOOP 2022

SCHOLARSHIPS

- ❖ Funding by ACM PLMW@ICFP to attend ICFP'19 in Berlin, Germany
- ❖ Pilot Scheme Funding by the University of Hong Kong to attend OPLSS'19 in Eugene, USA
- ❖ Partial funding by OPLSS'19 to attend OPLSS'19 in Eugene, USA
- ❖ University Postgraduate Fellowship for 4-years PhD by the University of Hong Kong (2018 – 2022)
- ❖ Merit Base Scholarship three times by PUCIT, University of the Punjab

KEY ACHIEVEMENTS

- ❖ Promoted to Senior Software Engineer and Team Lead after two years of professional experience at Platalytics Inc., Lahore, Pakistan which normally takes three years.
- ❖ 2nd position (2/191) in BS Computer Science (2011 – 2015) at PUCIT, University of the Punjab, Lahore, Pakistan.
- ❖ 2nd position in Belal Hashmi programming competition SOFTEC'15 ACM-ICPC held in FAST-NU Lahore (Team Code:Threads).
- ❖ 2nd position in speed programming competition SOFTEC'14 ACM-ICPC held in FAST-NU Lahore (Team Code:Threads).
- ❖ 2nd position in speed programming competition APTA'14 held in GCU Lahore (Team Code:Threads).

EXTRA CURRICULAR / SOCIAL ACTIVITIES

- ❖ Postgraduate Student's Representative at the Department of Computer Science, University of Hong Kong (Sep 2018 – Aug 2019)
- ❖ Member of Postgraduate Task Force on Accommodation Needs, Graduate School, University of Hong Kong (Oct 2018 – Aug 2019).
- ❖ Founding Member at Ghoomo Phiro Pakistan Adventure Club (Dec 2014 – Aug 2018)
 - Ghoomo Phiro Pakistan Adventure Club was established in 2014. The main idea behind this company is to promote tourism and cultural diversity in Pakistan. We successfully have organized many tours in the northern areas of Pakistan. Also we got funding from provincial Government of Khyber Pakhtunkhwa, Pakistan, for this purpose and we are working successfully as a team to achieve our objective.
- ❖ Class Representative at PUCIT, University of the Punjab (Oct 2013 – July 2015)
- ❖ Advisory Board Member at Blood Donor Society, PUCIT (Jan 2015 – July 2015)
- ❖ Vice president and Treasurer at Literary society, PUCIT (Oct 2012 – July 2015)
- ❖ BCSF11M cricket team player (Right Arm Medium Fast Bowler)